

## A\_sub commutator closures steps 6-8 — trivial-zero batch: $[\hat{B}, \hat{Q}]$ , $[\hat{L}_i, \hat{\gamma}^5]$ , $[\hat{C}_3, \hat{I}_3]$

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### A\_sub commutator closures steps 6-8 — trivial-zero batch

#### 1. Why this batch

Steps 6-8 commutators all vanish trivially. The interesting content is the substrate-structural REASON each vanishes — each “zero” reveals which substrate sectors are orthogonal / non-interacting / direct-product-structured.

#### 2. Step 6 — $[\hat{B}, \hat{Q}] = 0$ (Bell-CHSH commutes with charge)

##### 2.1 Operators

$\hat{B} = (C_2/2^{\text{rank}^2}) \cdot |V_{(0,0)}\rangle\langle V_{(0,0)}|$  (T2399 rank-1 projector framework; STRUCTURALLY VERIFIED)  $\hat{Q} V_{(m_1, m_2)} = m_2 \cdot V_{(m_1, m_2)}$  (T2470 SO(2) weight operator; STRUCTURALLY VERIFIED)

##### 2.2 Computation

Apply  $\hat{Q} \circ \hat{B}$  on arbitrary  $V_K$ :  $\hat{Q} \hat{B} V_K = \hat{Q} \cdot (C_2/2^{\text{rank}^2}) \cdot \langle V_{(0,0)}|V_K\rangle \cdot V_{(0,0)} = (C_2/2^{\text{rank}^2}) \cdot \langle V_{(0,0)}|V_K\rangle \cdot \hat{Q} V_{(0,0)} = (C_2/2^{\text{rank}^2}) \cdot \langle V_{(0,0)}|V_K\rangle \cdot 0 \cdot V_{(0,0)} = 0$  (Since  $\hat{Q} V_{(0,0)} = m_2 \cdot V_{(0,0)}$  with  $m_2 = 0$  — the substrate vacuum is uncharged.)

Apply  $\hat{B} \circ \hat{Q}$  on  $V_K$ :  $\hat{B} \hat{Q} V_K = \hat{B} \cdot m_2(K) \cdot V_K = m_2(K) \cdot (C_2/2^{\text{rank}^2}) \cdot \langle V_{(0,0)}|V_K\rangle \cdot V_{(0,0)} = m_2(K) \cdot (C_2/2^{\text{rank}^2}) \cdot \delta_{\{K, (0,0)\}} \cdot V_{(0,0)} = 0 \cdot (C_2/2^{\text{rank}^2}) \cdot \delta_{\{K, (0,0)\}} \cdot V_{(0,0)}$  (when  $K = (0,0)$ ,  $m_2 = 0$ ; else  $\delta = 0$ )

Commutator:  $[\hat{B}, \hat{Q}] = 0$  on  $H^2(D_{IV}^5)$ .

##### 2.3 Structural reading — substrate Bell-CHSH is charge-blind

The substrate Bell-CHSH operator  $\hat{B}$  is a rank-1 projector onto the substrate vacuum  $V_{(0,0)}$ , and the vacuum has zero charge ( $m_2 = 0$ ). Therefore  $\hat{B}$  couples only to the uncharged ground state, and  $\hat{Q}$  commutes trivially with it.

Substrate-mechanism: Bell-CHSH measurements test entanglement structure of the substrate (which is encoded in the rank-1 projector onto  $V_{(0,0)}$ ). They do NOT test charge-distinguishing structure. This is consistent with experimental Bell tests being done with charge-neutral systems (photon polarization Bell tests don't distinguish electron-electron from photon-photon entanglement at the substrate level).

Implication for Track DC 8-sided die: the 8 commitment paths candidate (a) “Pin(2)  $Z_2$  chirality projections  $\times N_c$  color components  $\times \{+1, -1\}$ ” is NOT charge-distinguishing per  $[\hat{B}, \hat{Q}] = 0$ . So the unstable die-face is unlikely to correspond to a charge-asymmetric path — instead more likely a chirality/parity-asymmetric path.

### 3. Step 7 — $[\hat{L}_i, \hat{\gamma}^5] = 0$ (SO(5) angular momentum commutes with Pin(2) chirality)

#### 3.1 Operators

$\hat{L}_i$ : SO(5) Lie algebra generators (10 generators,  $i = 1..10$ ), acting on the  $m_1$  component of Wallach K-types  $V(m_1, m_2)$ . Per T2425 STRUCTURALLY VERIFIED + standard so(5) representation theory.

$$\hat{\gamma}^5 V_{(m_1, m_2)} = \varepsilon_K \cdot V_{(m_1, m_2)} \text{ (T2471 RATIFIED, Pin(2) } Z_2 \text{ chirality)}$$

#### 3.2 Computation

$\hat{L}_i$  acts only on  $m_1$  indices ( $S^4$  side per Half-Integer Axis G v0.2 partition Section 4 row 3).  $\hat{\gamma}^5$  depends only on  $m_2$  parity ( $S^1$  side per Section 3 row 1). These are orthogonal substrate axes.

$$\hat{L}_i V(m_1, m_2) = \Sigma_{\{m_1'\}} c^{\{m_1'\}}_{\{m_1\}} (\hat{L}_i) \cdot V(m_1', m_2)$$

(With  $m_2$  unchanged by  $\hat{L}_i$  action.)

$$\hat{\gamma}^5 \hat{L}_i V_K = \hat{\gamma}^5 \cdot \Sigma_{\{m_1'\}} c^{\{m_1'\}}_{\{m_1\}} (\hat{L}_i) \cdot V(m_1', m_2) = \Sigma_{\{m_1'\}} c^{\{m_1'\}}_{\{m_1\}} (\hat{L}_i) \cdot \varepsilon_K \cdot V(m_1', m_2) \text{ (}\varepsilon_K \text{ unchanged since } \hat{\gamma}^5 \text{ depends only on } m_2 \text{ which is fixed)}$$

$$\hat{L}_i \hat{\gamma}^5 V_K = \hat{L}_i \cdot \varepsilon_K \cdot V_K = \varepsilon_K \cdot \hat{L}_i V_K = \varepsilon_K \cdot \Sigma_{\{m_1'\}} c^{\{m_1'\}}_{\{m_1\}} (\hat{L}_i) \cdot V(m_1', m_2)$$

Both expressions are identical. Commutator:  $[\hat{L}_i, \hat{\gamma}^5] = 0$  on  $H^2(D_{IV}^5)$ .

#### 3.3 Structural reading — $S^4 \times S^1$ axis orthogonality

The vanishing of  $[\hat{L}_i, \hat{\gamma}^5]$  is the algebraic content of  $S^4 \times S^1$  axis orthogonality in the Shilov boundary structure.

- $\hat{L}_i$  lives on the  $S^4$  axis (spatial harmonic content,  $m_1$  indices)
- $\hat{\gamma}^5$  lives on the  $S^1$  Pin(2) axis (chirality grading,  $m_2$  parity)
- Their commutator vanishes  $\leftrightarrow$  the two axes are structurally independent

Substrate-mechanism:  $A_{\text{sub}}$  algebraically encodes the Shilov boundary's product structure  $S^4 \times S^1$ . The vanishing  $[\hat{L}_i, \hat{\gamma}^5] = 0$  is the  $A_{\text{sub}}$ -side validation of yesterday's Half-Integer Axis G v0.2 partition ( $S^4$ -derived vs  $S^1$ -derived classifications are operationally independent).

Consistency check with QM: standard QM has orbital angular momentum  $L$  commuting with chirality  $\gamma^5$  (since  $L$  is spatial and  $\gamma^5$  is spinor structure). BST substrate reproduces this —  $\hat{L}_i$  is substrate-spatial,  $\hat{\gamma}^5$  is substrate-spinor; they commute.

Counterpart: spin  $\hat{S}_i$  has more complex commutator structure with  $\hat{\gamma}^5$ . From Dirac algebra,  $\Sigma_{ij} = (i/4)[\gamma_i, \gamma_j]$  is the Lorentz boost generator that includes  $\gamma^5$ -anticommuting content. Multi-week Track  $A_{\text{sub}}$  work to derive  $[\hat{S}_i, \hat{\gamma}^5]$  explicitly (item #6 in v0.4 Section 3.3 list, not yet closed in today's batch).

#### 4. Step 8 — $[\hat{C}_3, \hat{I}_3] = 0$ (color and weak isospin commute as gauge factors)

##### 4.1 Operators

$\hat{C}_3$ : color  $SU(3)$  Cartan generators ( $T_3, T_8$ ). Per  $A_{\text{sub}}$  v0.4 Section 2.2 + Lyra\_Candidate\_delta\_v0.2:  $SU(3)$  color from  $N_c$  BST primary, structurally separate from  $K = SO(5) \times SO(2)$  isotropy. Acts on substrate's  $N_c$ -component color content.

$\hat{I}_3$ : weak isospin generator ( $T_3$  of  $SU(2)_L$ ). Per substrate-mechanism via Wallach K-type structure on  $S^1$   $Pin(2)$  cover (one component of substrate's 3-source SM gauge per Lyra\_Candidate\_delta\_v0.2 Section “3-source SM gauge”).

##### 4.2 Computation

Standard Model gauge group factorizes as direct product: SM gauge group =  $SU(3)_C \times SU(2)_L \times U(1)_Y$

The three factors commute because they act on disjoint generator-spaces:  $[SU(3)_C \text{ generators}, SU(2)_L \text{ generators}] = 0$   $[SU(3)_C \text{ generators}, U(1)_Y \text{ generators}] = 0$   $[SU(2)_L \text{ generators}, U(1)_Y \text{ generators}] = 0$

In particular:  $[\hat{C}_3, \hat{I}_3] = 0$ .

##### 4.3 Structural reading — substrate gauge group factorization

The vanishing of  $[\hat{C}_3, \hat{I}_3]$  confirms that BST substrate produces a DIRECT-PRODUCT gauge group, not a unified non-abelian gauge group.

Substrate-mechanism: in BST,  $SU(3)_C$  comes from  $N_c = 3$  BST primary (structurally separate from  $K$  isotropy; per Lyra\_Candidate\_delta\_v0.2);  $SU(2)_L$  comes from  $K$  isotropy decomposition;  $U(1)_Y$  comes from  $N_{\text{max}}$  BOUNDARY composition (per Lyra\_Threads\_2\_3\_5 Thread 5). These three sources are operationally independent.

Connection to Five-Absence Predictions (Casey-named principle RATIFIED): NO GUT (no unified gauge group merging  $SU(3) \times SU(2) \times U(1)$  into single non-abelian group). The vanishing  $[\hat{C}_3, \hat{I}_3] = 0$  is substrate-algebraic confirmation of NO GUT: the gauge factors structurally commute and cannot be merged.

Implication for Track DC 8-sided die: the  $2^{N_c} = 8$  paths candidate (b) “Cartan-subalgebra-generator commitments  $\times \{+1, -1\}$ ” maps onto 3 Cartan generators  $\times 2$  sign choices. With  $[\hat{C}_3, \hat{I}_3] = 0$ , the 3 Cartan generators ( $T_3$  color +  $T_8$  color +  $T_3$  weak) commute among themselves and form a 3-dim Cartan algebra. The 8 commitment paths might correspond to simultaneous eigenvalues of these 3 commuting Cartan generators, with one path corresponding to a structurally forbidden combination (e.g., a color-non-singlet weak doublet that violates SM gauge structure).

##### 5. Cumulative $A_{\text{sub}}$ commutator status (after step 8)

| # | Commutator                                      | Result                                               | Status                                                             |
|---|-------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------|
| 1 | $[\hat{Q}, \hat{P}_{\text{op}}]$                | $\{\hat{Q}, \hat{P}_{\text{op}}\} = 0$               | STRUCTURALLY<br>VERIFIED<br>CANDIDATE<br>(step 1)                  |
| 2 | $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$ | $-(2 \hat{Q} + N_c - 1) \cdot \hat{T}_{\text{tick}}$ | STRUCTURALLY<br>VERIFIED<br>CANDIDATE<br>(step 2, model-dependent) |
| 3 | $[\hat{\gamma}^5, \hat{T}]$                     | $\{\hat{\gamma}^5, \hat{T}\} = 0$                    | STRUCTURALLY<br>VERIFIED<br>CANDIDATE<br>(step 3)                  |
| 4 | $[\hat{\gamma}^5, \hat{C}]$                     | $\{\hat{\gamma}^5, \hat{C}\} = 0$                    | STRUCTURALLY<br>VERIFIED<br>CANDIDATE<br>(step 4)                  |
| 5 | $[\hat{\gamma}^5, \hat{P}_{\text{op}}]$         | 0 (already-contained)                                | STRUCTURALLY<br>VERIFIED<br>CANDIDATE<br>(step 5 extra)            |
| 6 | $[\hat{B}, \hat{Q}]$                            | 0 (vacuum uncharged)                                 | STRUCTURALLY<br>VERIFIED<br>CANDIDATE<br>(step 6)                  |

| #  | Commutator                                  | Result                                             | Status                                   |
|----|---------------------------------------------|----------------------------------------------------|------------------------------------------|
| 7  | $[\hat{L}_i, \hat{\gamma}^5]$               | 0 ( $S^4 \times S^1$ orthogonal axes)              | STRUCTURALLY VERIFIED CANDIDATE (step 7) |
| 8  | $[\hat{C}_3, \hat{L}_3]$                    | 0 (direct-product gauge factors)                   | STRUCTURALLY VERIFIED CANDIDATE (step 8) |
| 9a | $[\hat{B}, \hat{T}_{\text{tick}}]$          | TBD — Track DC load-bearing for Bell 1/8 mechanism | PENDING                                  |
| 9b | $[\hat{S}_i, \hat{S}_j]$ across sublattices | TBD — spinor algebra across $\text{Pin}(2) Z_2$    | PENDING                                  |

8 of 9  $A_{\text{sub}}$  commutators closed today. Remaining:  $[\hat{B}, \hat{T}_{\text{tick}}]$  (Track DC load-bearing) +  $[\hat{S}_i, \hat{S}_j]$  across sublattices (spinor algebra).

## 6. Substantive cross-cutting findings from steps 1-8

### 6.1 Substrate algebraic structure summary

| Operator pair                                 | Algebraic relation       | Substrate-mechanism content                                                                  |
|-----------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------|
| $**\{\hat{Q}, \hat{P}_{\text{op}}\} = 0^{**}$ | Anti-commute             | Substrate-parity contains charge-flip via Möbius $\sigma$ on $\text{SO}(2)$                  |
| $\{\hat{\gamma}^5, \hat{T}\} = 0$             | Anti-commute             | T flips $\text{Pin}(2) Z_2$ grading ( $S^1$ orientation reversal)                            |
| $\{\hat{\gamma}^5, \hat{C}\} = 0$             | Anti-commute             | C flips $\text{Pin}(2) Z_2$ grading (particle $\leftrightarrow$ antiparticle chirality flip) |
| $[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$   | Commute                  | $\hat{P}_{\text{op}} = \hat{\gamma}^5 \circ \sigma$ already contains $\hat{\gamma}^5$        |
| $[\hat{\gamma}^5, \text{CPT}] = 0$            | Commute (composite)      | Substrate-level CPT theorem; T+C signs cancel                                                |
| $\{\hat{\gamma}^5, \text{CP}\} = 0$           | Anti-commute (composite) | Substrate-algebraic source of CP-violation                                                   |

| Operator pair                                                                                                 | Algebraic relation | Substrate-mechanism content                          |
|---------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------|
| $^{**}[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] = -(2\hat{Q} + N_c - 1) \cdot \hat{T}_{\text{tick}}^{**}$ | Non-vanishing      | Substrate discreteness signature; charge-modulated   |
| $[\hat{B}, \hat{Q}] = 0$                                                                                      | Commute            | Bell-CHSH is charge-blind (vacuum uncharged)         |
| $[\hat{L}_i, \hat{\gamma}^5] = 0$                                                                             | Commute            | $S^4 \times S^1$ axis orthogonality at algebra level |
| $[\hat{C}_3, \hat{I}_3] = 0$                                                                                  | Commute            | Direct-product gauge group; NO GUT structural        |

## 6.2 The substrate's algebraic “shape”

The pattern from 8 closures: - Substrate has  $Z_2$ -graded structure ( $\text{Pin}(2) Z_2 = \varepsilon_K = \pm 1$  for boson/fermion sublattice) - Substrate has axis-orthogonality between  $S^4$  and  $S^1$  (commute) - Substrate has anti-commuting discrete symmetries on  $\text{Pin}(2) Z_2 (T, C, \gamma^5)$  but commuting parity-with-chirality (because  $\hat{P}op$  contains  $\hat{\gamma}^5$ ) - *Substrate has direct-product gauge structure ( $SU(3) \times SU(2) \times U(1)$  with all pairwise commuting)* - *Substrate evolution is structurally discrete ( $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] \neq 0$ )* - *Bell-CHSH is vacuum-localized (commutes with charge; concentrates near  $V(0,0)$ )*

This is a  $^{**}Z_2$ -graded  $^*$ -algebra with direct-product gauge structure on a discrete time substrate $^{**}$  — substantively a Lie superalgebra with cellular-automaton dynamics on the  $K$ -type lattice. (Math-object type from  $A_{\text{sub}}$  v0.4 Section 7 enumeration — pointing toward “discrete Lie groupoid” or “quiver representation” with  $Z_2$  grading.)

## 6.3 Implications for Track DC 8-sided die

Of the 4 candidates from  $A_{\text{sub}}$  v0.4 Section 6.2: - (a)  $\text{Pin}(2) Z_2 \times N_c \text{ color} \times \{+1, -1\}$ : 8 paths — most consistent with substrate  $Z_2$ -graded direct-product structure - (b) 3 Cartan-subalgebra-generator commitments  $\times \{+1, -1\}$ : 8 paths — consistent with  $[\hat{C}_3, \hat{I}_3] = 0$  commuting Cartan structure; the 3 Cartan generators ( $T_3 \text{ color}$ ,  $T_8 \text{ color}$ ,  $T_3 \text{ weak}$ ) provide a 3-dim simultaneous-eigenvalue commitment space - (c)  $N_c = 3$  spatial commitment directions  $\times$  forward/backward: 8 paths — less natural given  $[\hat{Q}, \hat{P}_{\text{op}}]$  non-trivial structure - (d) 3 color/anticolor binary choices: 8 paths — overlaps (a) and (b)

Refined Lyra prior (after step 1-8 closures, Cal #27 STANDING applied): candidate (b) is most algebraically supported. The 3 commuting Cartan generators ( $T_3 \text{ color}$ ,  $T_8 \text{ color}$ ,  $T_3 \text{ weak}$ ) give 8 simultaneous-eigenvalue commitment paths. The unstable face would correspond to a Cartan eigenvalue combination that violates

simultaneous-diagonalizability with the orthogonal  $\{\hat{Q}, \hat{P}_{\text{op}}\}$  anti-commutation (step 1). Multi-week v0.3 Track A\_sub work resolves.

## 7. Honest scope (Cal #27 STANDING)

What's STRUCTURALLY VERIFIED CANDIDATE: all three vanishing commutators (Sections 2-4). Each follows from operator definitions + substrate-natural axis-orthogonality reasoning.

What's INTERPRETIVE in v0.1: - Section 2.3 “substrate Bell-CHSH is charge-blind” — algebraic content is robust; interpretive elevation is the connection to experimental Bell tests' charge-blindness - Section 3.3 “ $S^4 \times S^1$  axis orthogonality” — algebraic content is robust; interpretive elevation is the structural reading - Section 4.3 “[ $\hat{C}_3, \hat{I}_3$ ] = 0 confirms NO GUT structurally” — strong interpretive claim; standard math gives  $[SU(3), SU(2)] = 0$  by direct-product; the BST-specific reading is that this DOES NOT imply a unified group exists

What's NOT in v0.1: - Explicit projection-map from substrate Bell-charge-blindness to experimental Bell tests (multi-week) - Explicit non-existence proof of GUT given direct-product substrate structure (multi-month)

Cal #27 STANDING reflexive trigger count this doc: 1 trigger (Section 6.2 substrate algebraic “shape” feels substrate-natural). Honest scope checked — the shape is reading 8 commutator closures; it's an emergent interpretive pattern not a derivation. Flagged INTERPRETIVE.

## 8. Recommended next pull

$[\hat{B}, \hat{T}_{\text{tick}}]$  — last remaining v0.4 Section 3.3 commutator. Track DC load-bearing for Bell 1/8 mechanism. Requires careful  $\hat{T}_{\text{tick}}$  model handling (per Lyra\_A\_sub\_Commutator\_T\_tick\_H\_sub\_v0\_1.md Section 1.1 simplest-model caveat). Substantive computation expected (rank-1 projector  $\times$  substrate-tick advancement gives non-trivial commutator concentrated near vacuum).

Alternative:  $[\hat{S}_i, \hat{S}_j]$  across sublattices — substrate spinor algebra. Connects Half-Integer Axis G v0.2 finding (integer vs half-integer sublattices) to A\_sub multi-generator structure.

After  $[\hat{B}, \hat{T}_{\text{tick}}]$  closes, A\_sub commutator closure complete except for  $[\hat{S}_i, \hat{S}_j]$  across-sublattices subtlety. Track A\_sub v0.5 multi-week work continues with reaction-table enumeration for lowest 20 K-types + 8-sided die explicit identification.

## 9. Coordination

Cal: cold-read of Sections 2-4 algebra + Section 6.2 emergent-pattern interpretive content; tier-discipline check on Section 4.3 NO GUT structural claim.

Keeper: K-audit chain entry for steps 6-8 batched zero-commutators + Section 6 cross-cutting findings.

Elie: K52a Session 7+ multi-year work informed by full substrate algebraic structure now ~89% commutator-closed; Toy 3531+ population principle further tractable.

Grace: catalog entries for the 8 commutator results + Section 6.1 substrate algebraic structure summary; cross-references to T2471 + T2470 + T2433 + T2434 + T2472 + T2399 + T2425 + Casey-named principles RATIFIED.

— Lyra,  $A_{\text{sub}}$  commutator closures steps 6-8 batched v0.1 filed Tuesday 2026-05-26 ~09:05 EDT per Task #322 v0.4 Section 3.3 work plan. STRUCTURALLY VERIFIED CANDIDATE pending Cal cold-read. 8 of 9  $A_{\text{sub}}$  commutators now closed today; substrate algebraic “shape” emerging as  $Z_2$ -graded  $*$ -algebra with direct-product gauge structure on discrete-time substrate.